MIL-H-3117C(SHIPS)
1 February 1962
SUPERSEDING
MIL-H-3117B(SHIPS)
9 July 1953

MILITARY SPECIFICATION

HEATERS, CONVECTION, STEAM AND HOT WATER

1. SCOPE

1.1 This specification covers steam and hot water convection heaters with cabinets for use in heating spaces on Naval ships.

1.2 Classification. - The convection heaters shall be of the following classes and sizes as specified (see 6.1):

Class 1 - Carbon steel construction.
Class 2 - Nonmagnetic construction:
Size 1-1/2 - 1200 British thermal units per hour (BTU/HR) minimum capacity.
Size 2 - 1600 BTU/HR capacity.
Size 2-1/2 - 2000 BTU/HR capacity.
Size 3-1/2 - 3200 BTU/HR capacity.
Size 4-1/2 - 4500 BTU/HR capacity.
Size 6 - 6100 BTU/HR capacity.
Size 8 - 8400 BTU/HR capacity.
Size 11 - 11,000 BTU/HR capacity.
Size 15 - 14,500 BTU/HR capacity.
Size 20 - 20,000 BTU/HR capacity.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

MILITARY
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-Q-9858 - Quality Control System Requirements.
MIL-P-15328 - Primer, Pretreatment (Formula No. 117 for Metals).

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-278 - Welding and Allied Processes for Ships of the United States Navy.

DRAWINGS

BUREAU OF SHIPS
S3802-921624 - General Type Plan Convection Heater.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)
A415 - Hot Rolled Carbon Steel Sheets, Commercial Quality.
B88 - Seamless Copper Water Tube.
B260 - Brazing Filler Metal.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

COMMERCE DEPARTMENT
Commercial Standard
CS 140-47 - Testing and Rating Convectors.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Qualification. - The steam and hot water convection heater furnished under this specification shall

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be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Materials. - Materials used in the construction of the heaters shall be of high grade, free from defects, and suitable for the purpose intended.

3.2.1 Materials specified herein shall be of uniform finish, free of holes, scale, abrasions, laminations, strains, foreign matter and other surface defects, and shall be capable of being punched or formed or both without the creation of cracks.

3.2.2 The class 1 convection heater shall be made of steel equal to that of ASTM A-415 except that the heating element shall be made of copper and the damper pin and metal parts of the damper adjusting mechanism shall be made of a non-ferrous material or corrosion-resisting steel.

3.2.3 The class 2 convection heater shall be made of aluminum, brass, copper or corrosion-resisting steel which has a permeability of less than 2.0 after fabrication. The heating element shall be made of copper.

3.2.3.1 Parts of the heater constructed of aluminum or aluminum alloy shall not be used in contact with any copper bearing metal.

3.3 Design. - The heater shall be designed for use in both steam and hot water heating systems.

3.3.1 Configuration, physical dimensions, tolerances and capacities. -

3.3.1.1 The configuration, physical dimensions and tolerances of the heater shall conform to Drawing S3802-921624 for the sizes specified in table I.

3.3.1.2 The capacities shall be as shown in table I for the indicated sizes. The capacities of table I are predicated on 35 pounds per square inch gage (psig) steam pressure and 65° Fahrenheit (F.) inlet air temperature.

Table I - Sizes and capacities.

Sizes	Capacities, BTU/HR, minimum
1-1/2	1,200
2	1,600
2-1/2	2,000
3-1/2	3,200
4-1/2	4,500
6	6,100
8	8,400
11	11,000
15	14,500
20	20,000

3.3.2 Air inlets and outlets. - The heat output and outlet air temperature of the heater shall be controlled by means of an induced air inlet with damper located in the front panel of the heater, above the heating element. This induced air inlet shall be independent of and separate from the air inlet below the heating element. The heater cabinet shall therefore have two air inlets, one below the heating element and the other above the heating element, and an air outlet in the sloping top of the cabinet.

3.3.3 Induced air damper for temperature regulation. -

3.3.3.1 The induced air damper shall be hinged on the back of the front panel below the induced air opening, and shall be adjustable to any position from induced air inlet closed and the air passage above the heating element fully open, to induced air inlet fully open and air passage above the heating element closed. The induced air damper shall be designed and arranged to facilitate induction of a flow of air through the induced air inlet, when the induced air damper is in positions above the heating element, by means of the convection currents of heated air from the heating element.

3.3.3.2 The design and arrangement of the induced air damper and size of induced air inlet and air outlet shall be such that a reduction of at least 75 percent in heat output and at least 40 percent in outlet air temperature from maximum heat output and outlet air temperature, respectively, may be obtained by adjustment of the induced air damper. When fully closed, the induced air damper shall close the induced air inlet opening and when fully open the induced air damper shall close the air passage above the heating element.

3.3.3.3 The induced air damper shall be manually operated by means of a knob or handle located on the front of the cabinet. The knob or handle shall not project beyond the front of the heater by more than 1-1/4 inches. Means shall be provided to indicate direction of operation to raise and lower heat output.

3.3.4 Grilles. - The air outlet and induced air inlet shall be provided with grilles. The free area through the outlet grille shall be at least equal to the free area through the heating element, and the free area through the induced air inlet grille shall be at least 70 percent of the free area through the heating element.

3.3.5 Heating element. -

3.3.5.1 The heating element shall be of the extended surface type, consisting of copper tubing with a wall thickness not less than that required for type L of ASTM B-88 expanded into fins and having the ends brazed or welded into collars on the headers. The headers shall be made of brass or copper. Brazed joints of the heating element shall be made

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with silver brazing alloy or phosphorous copper in accordance with ASTM B-260.

3.3.5.2 Fins shall be not less than 0.012 inch thick, and shall be uniformly spaced. The relation of the total fin surface to the clear area through the heating element shall be such that the maximum temperature of the outlet air will not exceed 180°F. at the specified operating steam pressure when inlet air is at 65°F.

3.3.5.2.1 The fin surface of the heating element shall be of copper known commercially as "electrolytic tough pitch copper".

3.3.5.3 Suitable means shall be provided for venting air from the heating element. The construction and arrangement of the heating element shall permit the air vent being fitted at either end of the element.

3.3.5.4 When tested as specified in 4.6.2 the heating element shall show no leakage.

3.3.5.5 Supports.- Supports for the heating element shall be designed to withstand stresses, jars, vibrations incident to shipping, installation and operation and shall be designed to prevent bypassing of air between heating element and cabinet. A heat insulating gasket not less than one-sixteenth inch thick, of asbestos or other heat insulating material, shall be fitted between the heating element and the supports.

3.3.6 Knockouts, or other suitable means, shall be provided in the sides of the cabinet to permit a connection from the top of the heating element headers.

3.3.6.1 Two brass street ells that will line up with the knockouts when installed shall be furnished with each heater. The street ells shall be placed in a bag and attached to the heater.

3.3.7 Four spacers as shown on Drawing S3802-921624, 3/8-inch long and sized to fit mounting bolts shall be furnished, except where mounting strip is located on outside of cabinet in a manner to provide an air space 3/8-inch or more than back of cabinet and bulkhead on which heater is to be mounted. The spacers shall be placed in the bag with the street ells (see 3.3.6.1).

3.4 Welding and brazing.- The surfaces of all parts to be welded or brazed shall be free from rust, scale, paint, grease and other foreign matter. All welding and brazing shall be at least equal to that required by MIL-STD-278.

3.5 Protective coatings.-

3.5.1 Brass surfaces, copper surfaces, corrosion-resisting steel surfaces, and the operating parts of the damper adjusting mechanism shall not be painted.

3.5.2 Unless otherwise specified herein, steel surfaces shall be given a hot-dipped phosphate treatment or a cold-wash phosphate-chromate treatment after fabrication prior to applying one coat of primer conforming to TT-P-664.

3.5.3 Aluminum alloy surfaces shall receive a thorough wash and hot phosphoric or chromic acid treatment or a pretreatment coating of MIL-F-15328 prior to applying two coats of primer conforming to TT-P-664.

3.6 Fasteners.-

3.6.1 All bolts, nuts, screws and other such fastenings as may be used shall be of a corrosion-resisting material suitable for intended application. They shall be of a recognized commercial standard size, shape, and tolerances in order to insure correct mating of parts and convenience of interchangeability and replacement. The threads of all machine screws, bolts and nuts shall conform to Unified coarse-thread series.

3.7 Drawings.- Drawings delineating the equipment shall be furnished in accordance with class C of Specification MIL-D-963 and as specified herein. Drawings are not to be titled to show application to particular ships or purchase orders.

3.7.1 Assembly drawings.- Assembly drawings shall show clearly the details of design, construction and identify each part and its location. Parts shall be identified in accordance with the numbers used in the list of material. The drawing shall show overall dimensions, mounting dimensions, attachment and connection dimensions, and tolerances. Such additional information, not required herein, shall be included on the drawing to reflect compliance with the specification requirements for the equipment.

3.7.2 Certification data.- Certification data shall be furnished and shall conform to class B of Specification MIL-D-963, with approval by the bureau or agency concerned.

3.8 Identification plates.- Each heater shall be provided with an identification plate not to exceed 3 by 3 inches. The identification plate may be etched, engraved, or cast, using either monel, brass, corrosion-resisting steel or aluminum alloy (see 3.2.3.1) or may be of anodized aluminum photographic process. The identification plate shall contain the following information:

(a) Item name.
(b) Size (see 3.8.1).
(c) Federal stock number.
(d) Component identification number (CID).

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(e) Contract number.
(f) Manufacturer's name.
(g) Manufacturer's address

3.8.1 The letters "NM" to designate that unit is constructed of non-magnetic material shall be added after the size designation on class 2 convection heater.

3.8.2 Anodized aluminum photographic process. - Anodized aluminum plates shall have the photosensitive silver compounds imbedded within the oxide layer. After photographic processing the image shall be sealed in the oxide layer by suitable chemical treatment of the layer. Stamping of additional marking information shall not be permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The contractor shall provide and maintain a quality control system acceptable to the Government for the supplies covered by this specification. The system of quality control shall be in accordance with MIL-Q-9858.

4.2 Qualification tests[1/]. - Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. These tests shall consist of the tests specified in 4.2.2, 4.6.1 and 4.6.2.

4.2.1 Samples for qualification. - Samples for qualification shall consist of a complete range of sizes of either class. Qualification of one class shall constitute qualification of both classes.

4.2.2 Capacity range and outlet temperature tests. - Each size of convection heater shall be tested in accordance with Commercial Standard CS 140-47 except as follows:

(a) The steam supply at the heater shall be at a pressure not less than 33 psig nor more than 37 psig with a superheat of not less than 2° nor more than 5°F.
(b) The mean outlet air temperature shall not exceed 180°F. when the air inlet temperature is 65°F., when the damper is in position to close off the induced air grille in the front of the cabinet.
(c) Capacity and outlet air temperature of each heater shall be corrected to 35 psig steam pressure and 65°F. inlet air temperature.
(d) Capacity and outlet air temperature shall be determined with damper in at least four positions.
(e) Allowance for "heating effect" as defined in Commercial Standard CS 140-47 is not to be included in the capacity of the unit.

4.3 Qualification check test. - When design changes are approved for a particular size of heater which may alter the capacity or increase the outlet air temperature of the heater, tests specified in 4.2.2 shall be conducted on that size of heater on the first item produced under the contract or order.

4.4 Sampling for acceptance inspection. -

4.4.1 Lot. - All convection heaters of the same size and class offered for delivery at one time shall be considered a lot.

4.4.2 Sampling for examination and tests. - Sample convection heaters shall be selected at random from each lot in accordance with table II for the examination specified in 4.5 and tests specified in 4.6. The Acceptable Quality Level (AQL) shall be 1.5 percent.

4.5 Examination. - Each of the sample convection heaters selected in accordance with 4.4.2 shall be examined and measured to verify compliance with the requirements of this specification not involving tests. Examination shall be conducted as specified in table III. Any convection heater in the sample containing one or more defects shall be rejected and if the number of defective convection heaters in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.6 Tests. - Each of the sample convection heaters selected in accordance with 4.4.2 shall be tested as specified in 4.6.1 and 4.6.2. Any convection heater found to be defective shall be rejected and if the number of defective convection heaters in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.6.1 Damper operation. - The damper assembly of each sample convection heater shall be operated to see that all parts fit properly and operate freely. The damper shall be checked to see that it is true and fits closely against the induced air opening in the closed position.

[1/] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.3).

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Table II - Sampling for examination and tests

Number of heaters in an inspection lot	Number of heaters in sample	Acceptance number (defective)	Rejection number (defective)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 to 1,300	75	3	4
1,301 and over	110	4	5

Table III - Classification of defects in accordance with MIL-STD-105

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Size not as specified.
102	Incomplete, parts missing.
103	Materials defective, not non-magnetic (if required), as specified.
104	Fastening hardware (bolts and nuts) not of corrosion-resistant material.
105	Copper-bearing metal parts come in direct contact with aluminum parts.
106	Air inlets and outlets missing, not of required size, grilles missing, or inlets and outlets not located as specified.
107	Damper missing, damaged, or not as specified.
108	Failure to provide a means to prevent bypass of air between cabinet and heater element; heat insulating gasket missing, damaged, or not as specified.
109	Heating element defective, fins not uniformly spaced, or not as specified.
110	Means of venting air from heating element not provided.
111	Joints in heating element defective, not secure, or not as specified.
112	Damper adjusting mechanism defective, not provided with a means of indicating direction of operation to raise and lower heat, or mechanism not located as specified.
113	Overall dimensions exceed the allowable.
114	Mounting dimensions not as specified.
115	Thickness of walls of tubes less than permissible.
Minor:	
201	Protective coatings missing, incomplete, or not as specified.
202	Marking not permanent, illegible, or not as specified.
203	Spacers and connecting ells missing, damaged, or not as specified.

4.6.2 Leakage tests. - The heating element of each sample convection heater shall be tested with 200 psig air pressure while immersed in water, 200 psig steam pressure, or 300 psig hydrostatic pressure.

4.7 Possible test failures:

(a) Failure of damper to operate freely, or damper fails to fit closely against air opening when in the closed position.

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(b) Evidence of leakage when heating element is subjected to specified air, steam, or hydrostatic pressure.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Heaters shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.- Unless otherwise specified in the contract or order, the heaters shall be packed for shipment in a manner which will insure acceptance by the carrier and safe delivery to destination. Shipping containers or method of packing shall comply with the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.3 Where special preservation, packaging or packing is required because of particular shipment or storage requirements, detail requirements shall be specified (see 6.1).

5.4 Marking.-

5.4.1 Unless otherwise specified (see 6.1), each package shall be marked to show the stock number, nomenclature, size, contract number, name and address of manufacturer of the packaged heater. Any additional markings required shall be specified in the contract or order.

5.4.2 Unless otherwise specified (see 6.1), shipping containers shall be marked to show stock numbers, nomenclature, quantity and sizes of contents, gross weight and cube, contract number, and the complete address of the consignor and the consignee. Any additional markings required shall be specified in the contract or order.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of specification.
(b) Class and size required (see 1.2).
(c) Packaging and packing requirements, if other than specified in 5.1 and 5.2 and preservation, if required (see 5.3).
(d) Marking requirements if other than specified in 5.4.

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL 3117, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C. and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

6.3 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may be related thereto.

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